Structured Output and Schemas

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Structured Output and Schemas before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Iterative Prompting

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Prompt as a Contract

Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

A Distribution of Possibilities

Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Structured Output and Schemas trace and record the first step you can explain without help.

Use W2: Audit **type**s and **field**s for Structured **output**s and **validation**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Structured Output and Schemas explanation against the supplied facts before discussing it.

Use the worked case for Structured **output**s and **validation**. Proposed explanation: “Valid JSON means correct answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Validate new objects independently

Sources A approve room F and B register F capacity 9. Object X has supported,F,9,[A,B] with exactly the required **field**s. Y replaces 9 with "9". Z keeps integer 9 but names room G.

Trace the first failing layer for each and explain the difference.

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Structured Output and Schemas target to a trace and a justified conclusion.

> **Schema** requires title:text and pages:number. **Output** is {"title":"Cloud","pages":"80"}. Fix its structural error. Would the repaired object prove the book really has 80 pages?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Structured Output and Schemas case and show the deciding evidence.

Schema requires title text and pages number. **Output** is {"title":"River","pages":"80"}.

Diagnose the shape error.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A required **schema** has title as text and count as an integer. The **output** is {"title":"Seeds","count":"three"}. Identify the error and give a valid replacement.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 8 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Structured Output and Schemas adds to Verified Study Buddy and keep the evidence.

Require answer, source snippet, citation, uncertainty, and next-question **field**s.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Schema requires title:text and pages:number. **Output** is {"title":"Cloud","pages":"80"}. Fix its structural error. Would the repaired object prove the book really has 80 pages?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Distinguish the layers

> Distinguish the layers. Apply the supplied Structured Output and Schemas facts and keep your reasoning.

Explain what parsing, **schema** checks and source support each establish.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Audit types and fields

> Audit types and fields. Apply the supplied Structured Output and Schemas facts and keep your reasoning.

Why do string "12",boolean true,missing capacity and an extra action **field** violate this contract?

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Check a citation-like ID

Check a citation-like ID. Apply the supplied Structured Output and Schemas facts and keep your reasoning.

Sources P,Q support B capacity 12. A **schema**-valid object cites P,Q but gives 99. Which layer rejects it?

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Validate new objects independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Validate new objects independently. Apply the supplied Structured Output and Schemas facts and keep your reasoning.

Sources A approve room F and B register F capacity 9. Object X has supported,F,9,[A,B] with exactly the required **field**s. Y replaces 9 with "9". Z keeps integer 9 but names room G.

Trace the first failing layer for each and explain the difference.

My reasoning and evidence

W5 Keep permissions separate

> Keep permissions separate. Apply the supplied Structured Output and Schemas facts and keep your reasoning.

A fully valid supported reporting object is returned. May the application automatically book the room? Explain.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Structured Output and Schemas

> A fictional Study Buddy returns a machine-readable book fact. Explain the labelled step before checking the changed condition.

A fictional Study Buddy returns a machine-readable book fact.

Schema: exactly title text, pages integer, source_id text. Approved record R: River Notes, 64 pages.

X={"title":"River Notes","pages":64,"source_id":"R"}; Y uses pages:"64"; Z uses pages:46 with otherwise identical **field**s.

AI ACADEMY / AI-3/8

Structured Output and Schemas

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Parse

Each shown object is valid JSON syntax.



2. Schema check

Y fails because its pages value is text, not an integer.



3. Meaning check

Z fits field types but contradicts R's page count.



4. Accept X

X passes syntax, declared schema and source comparison.

CHANGE ONE THING

Add an extra field "rating":5 to X under the exactly-fields schema.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which object demonstrates that valid JSON and valid **type**s are insufficient for truth?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Validate Study Buddy **output**s before another program consumes them; retain source IDs and separate syntax/**type**/support failures.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Validate Study Buddy **output**s before another program consumes them; retain source IDs and separate syntax/**type**/support failures.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy returns a machine-readable book fact.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? X passes syntax, declared **schema** and source comparison.

> 1. Meaning check

2. Accept X

3. **Schema** check

4. Parse

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Accept X X passes syntax, declared **schema** and source comparison. Parsing establishes that text forms a structured object.

Schema checks establish required **field**s and **type**s. **Semantic validation** compares their meaning with the approved source. These are different layers: a parseable, well-shaped object can still contain a false fact.

Report the first failing layer without treating format success as factual success.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The changed object fails **schema** despite keeping the correct page count.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The changed object fails **schema** despite keeping the correct page count. Exactly means additional **field**s are not allowed in this contract.

Repair a missing building block

Review Iterative Prompting (ai-3/7). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: What **evidence** supports improvement here, and what remains unproven?.

Return to this week and explain what you changed.

Review Prompt as a Contract (ai-3/5). Goal: Write a prompt specification with observable response criteria. Short practice: Why should the tool not fill melting from memory even if it knows a common definition?.

Return to this week and explain what you changed.

Review A Distribution of Possibilities (ai-3/2). Goal: Check whether a next-token table is a valid probability distribution. Short practice: Why is 0.625 not a broken probability sum?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

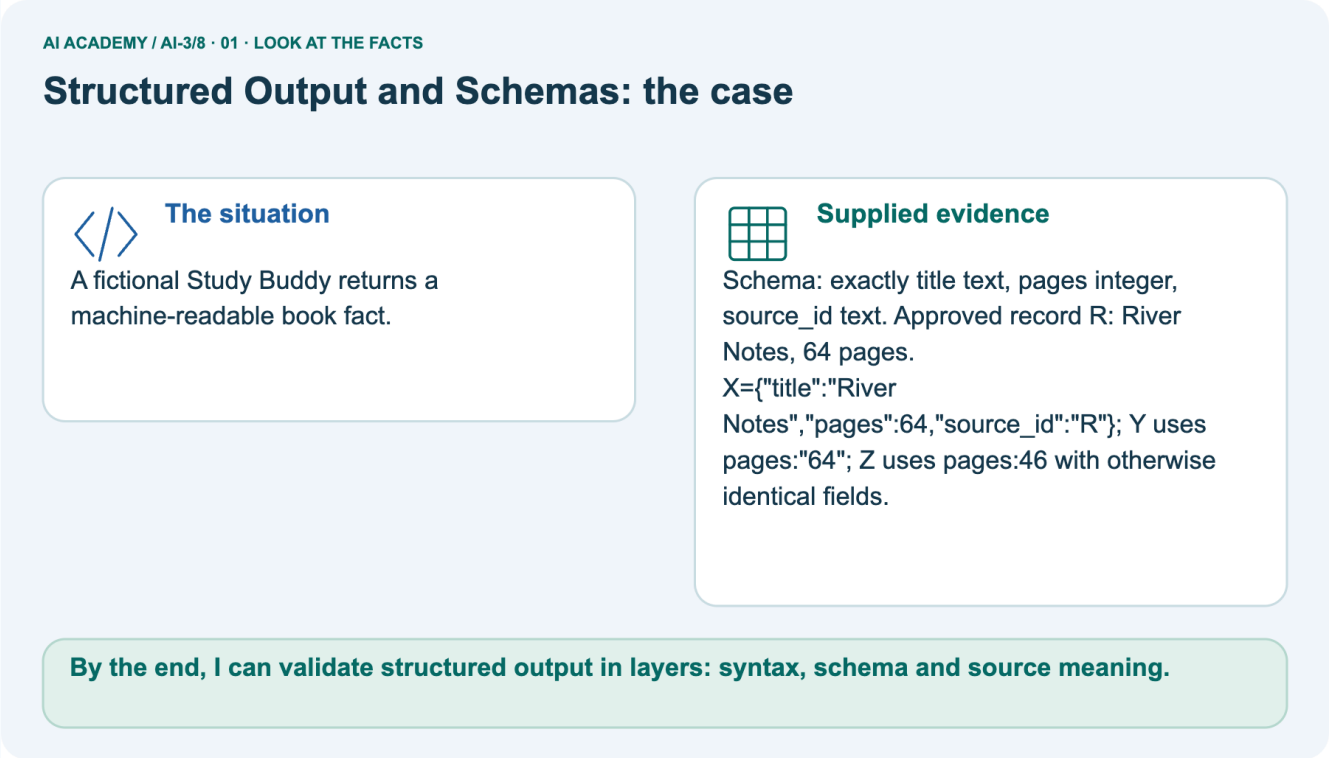
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Structured Output and Schemas picture from its labels, then record a relevant limit.

01. Structured Output and Schemas: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

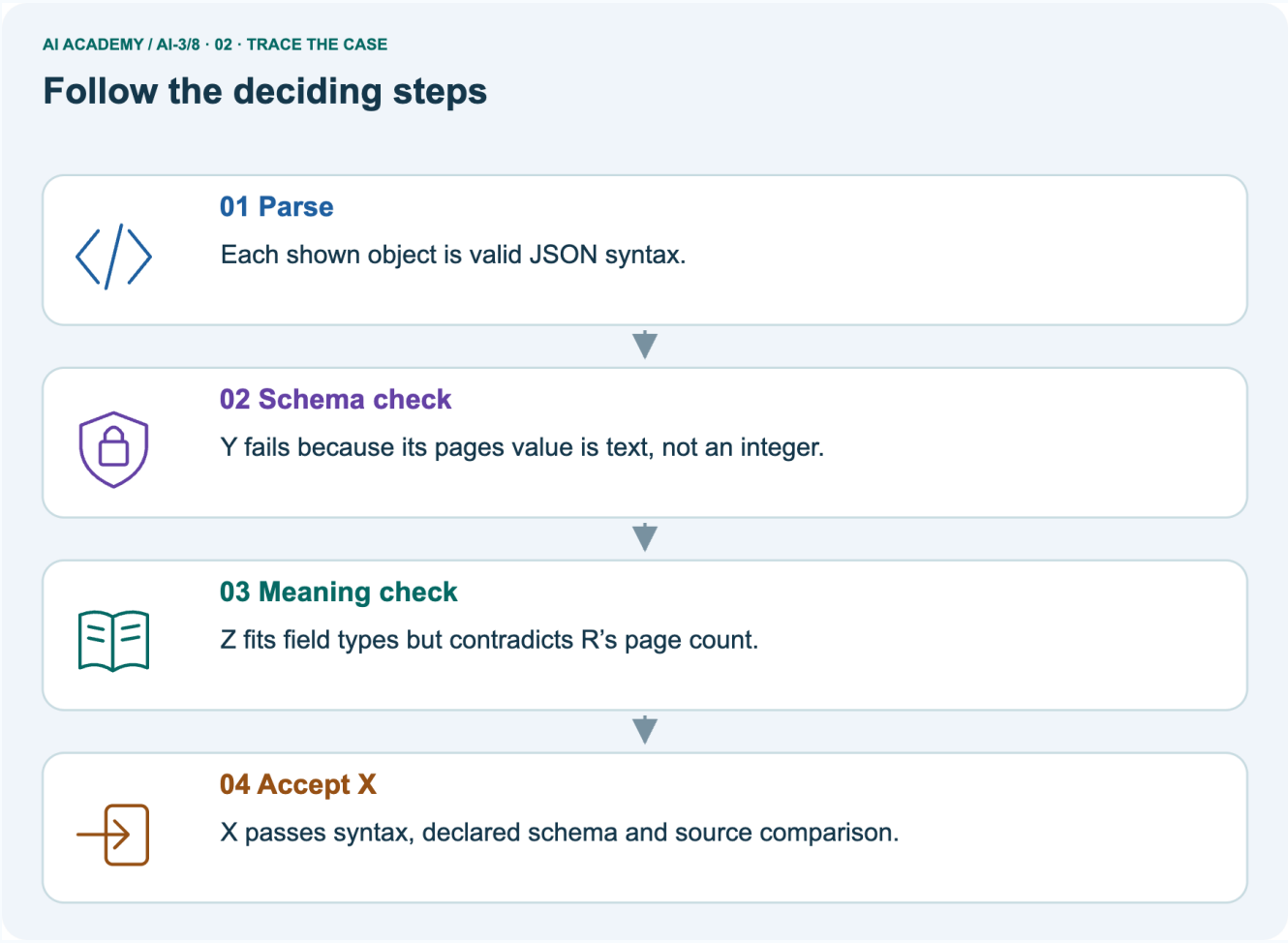
A fictional Study Buddy returns a machine-readable book fact. Supplied facts: **Schema**: exactly title text, pages integer, source_id text. Approved record R: River Notes, 64 pages.

X={"title":"River Notes","pages":64,"source_id":"R"}; Y uses pages:"64"; Z uses pages:46 with otherwise identical **field**s.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



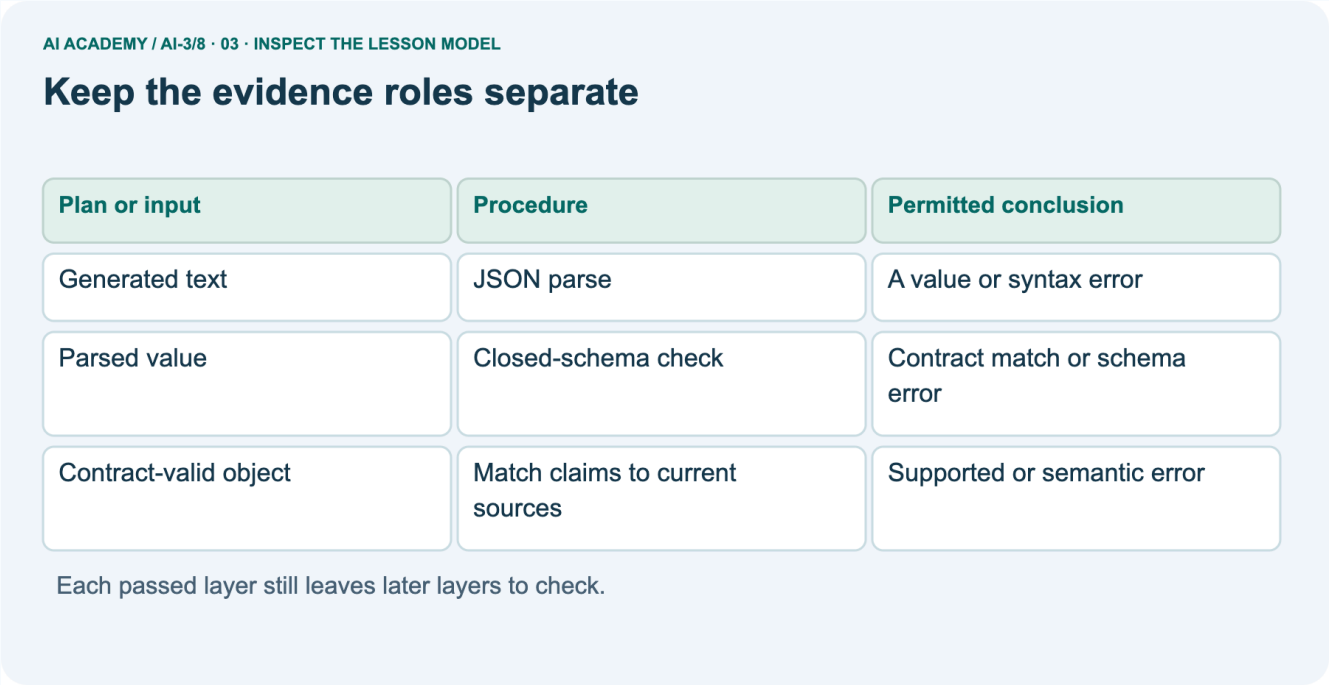
Follow the labelled picture with the matching explanation. Open the vector diagram

Parse: Each shown object is valid JSON syntax. → **Schema** check: Y fails because its pages value is text, not an integer. → Meaning check: Z fits **field type**s but contradicts R's page count. → Accept X: X passes syntax, declared **schema** and source comparison.

> Point to each numbered stage and explain which recorded **input** it uses.

> **My picture explanation, deciding evidence and one limit**

03. Keep the evidence roles separate



Follow the labelled picture with the matching explanation. Open the vector diagram

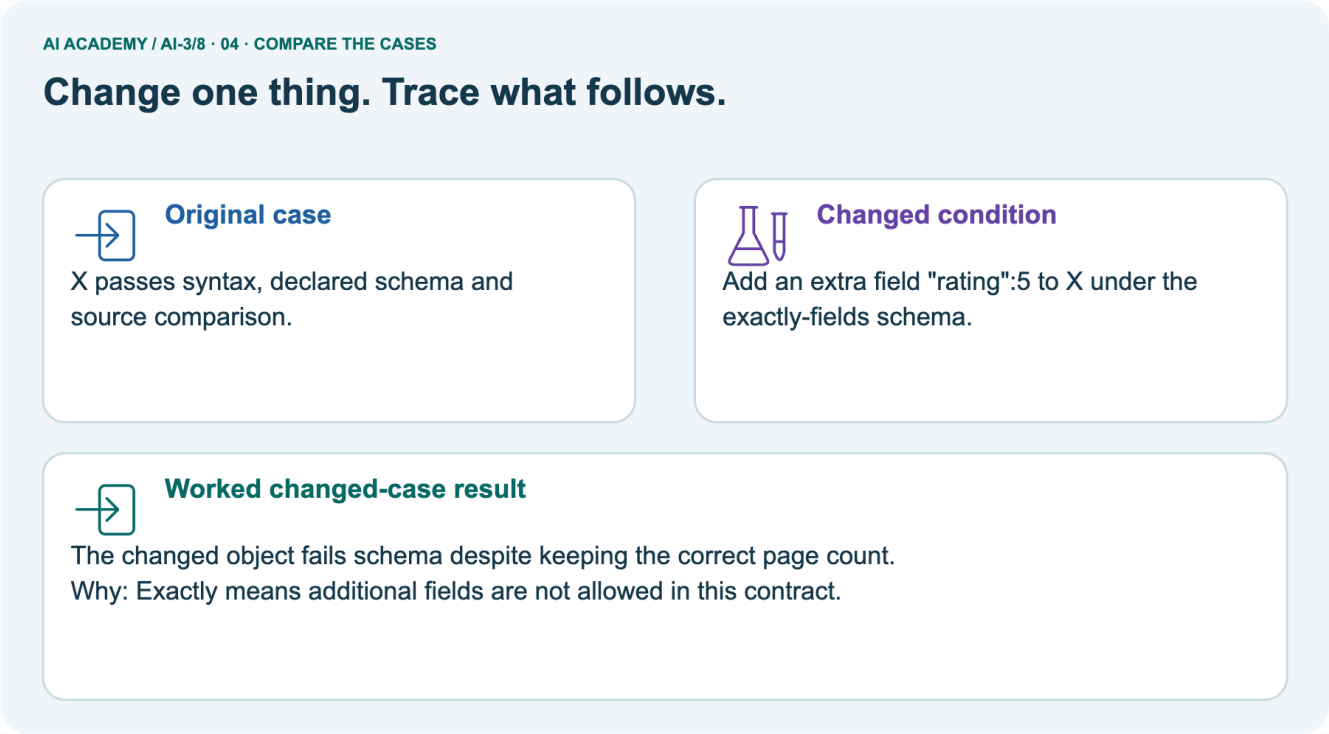
Keep the **evidence** roles separate. Generated text | JSON parse | A value or syntax error. Parsed value | Closed-**schema** check | Contract match or **schema** error.

Contract-valid object | Match claims to current sources | Supported or semantic error

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

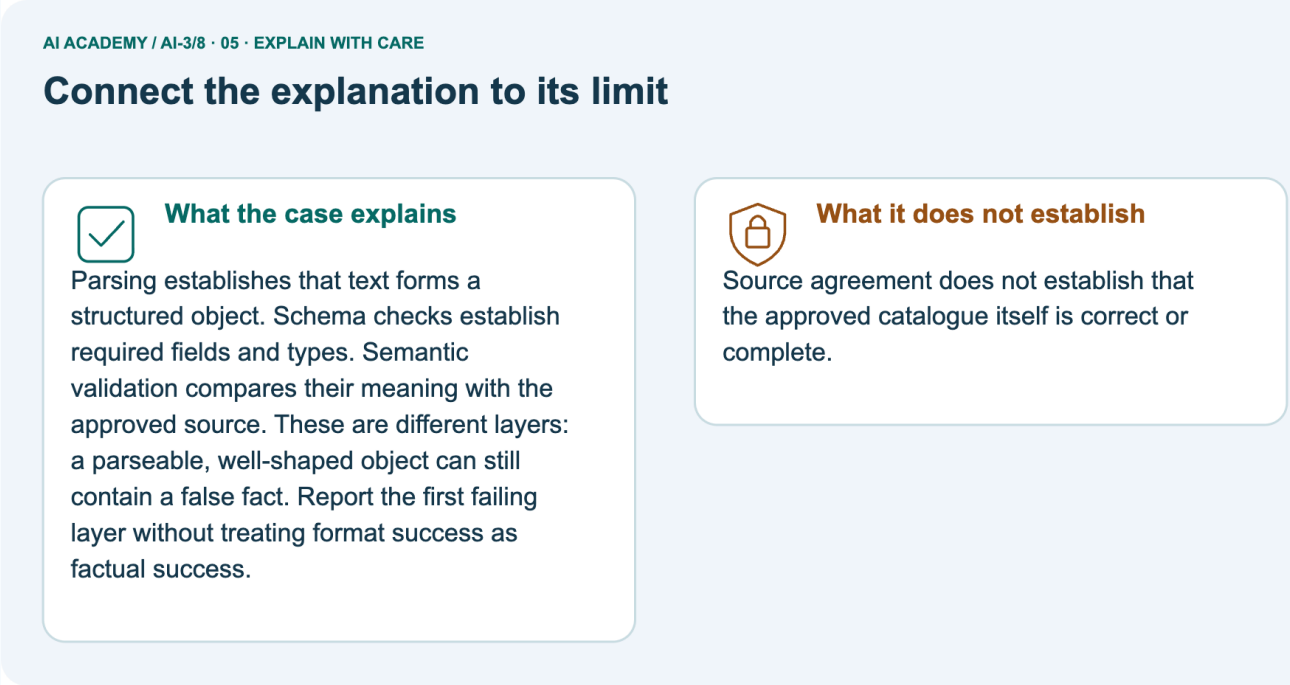
Original case: X passes syntax, declared **schema** and source comparison. Changed condition: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**.

Expected result: The changed object fails **schema** despite keeping the correct page count.
Reason: Exactly means additional **field**s are not allowed in this contract.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Parsing establishes that text forms a structured object. **Schema** checks establish required **field**s and **type**s. **Semantic validation** compares their meaning with the approved source.

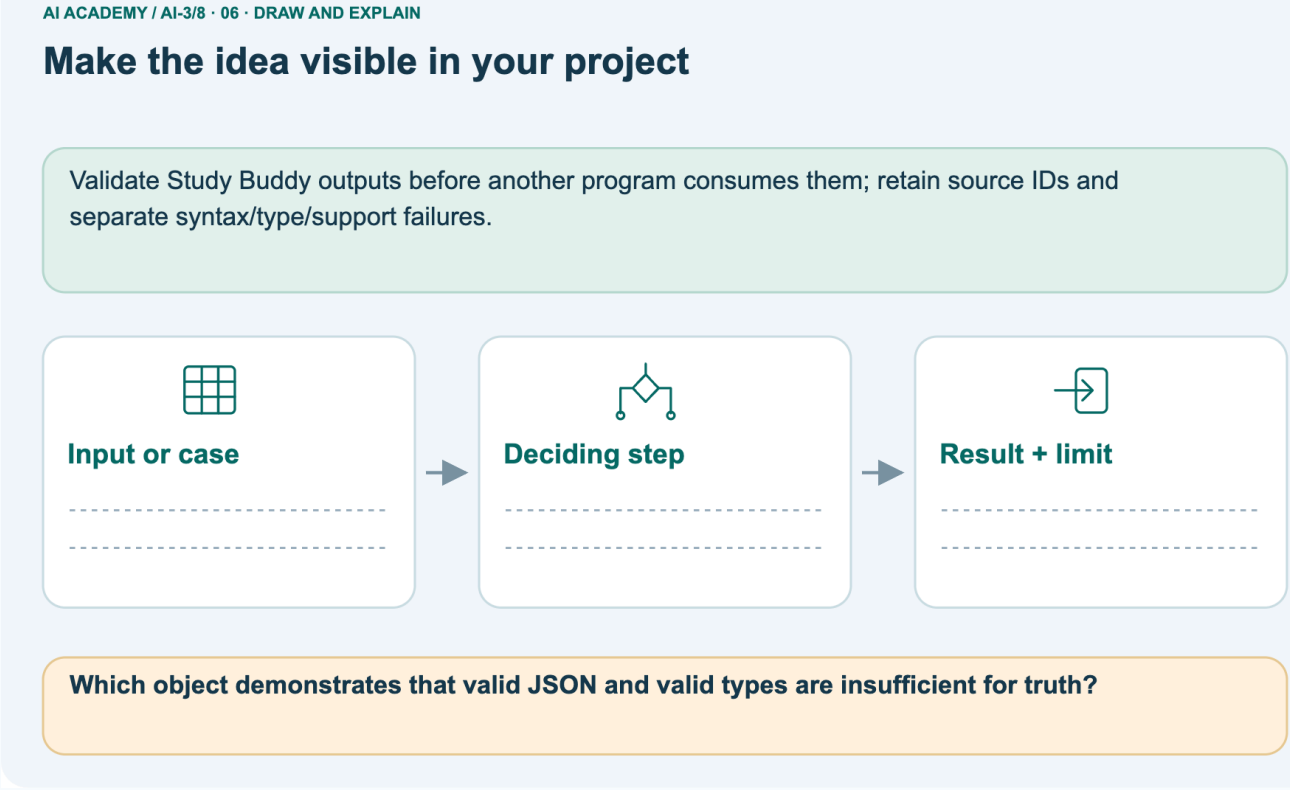
These are different layers: a parseable, well-shaped object can still contain a false fact. Report the first failing layer without treating format success as factual success.

Limit: Source agreement does not establish that the approved catalogue itself is correct or complete.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate Study Buddy **output**s before another program consumes them; retain source IDs and separate syntax/**type**/support failures. Draw your **input**, deciding step, result and limit.

Supported question: Which object demonstrates that valid JSON and valid **type**s are insufficient for truth?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.